For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY 9189/1

PAPER 1

JUNE 2016 SESSION 2 hours

Additional materials:
Answer paper
Data Booklet
Mathematical tables and/or electronic calculator

TIME: 2 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from Section C and **one** other question chosen from any section.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 9 printed pages and 3 blank pages.

Section A

Answer at least two questions from this section.

1 (a) (i) Define the term *ideal gas*.

(ii) Explain why noble gases behave more ideally than other gases.

[3]

(b) (i) A 185.00 g sample of fluorine gas and 4 moles of xenon were placed in a closed flask at 0 °C and 2.5 atm.

Deduce the volume of the gaseous mixture.

(ii) Addition of 23.00 g of lithium metal into the flask resulted in a violent reaction.

Write a chemical equation for the violent reaction.

(iii) Calculate the change in the

1. number of moles of fluorine,

2. total pressure.

[9]

[Total: 12]

2 **(a)** A Born-Haber Cycle for the formation of calcium sulphide is shown in **Fig. 2.1.**

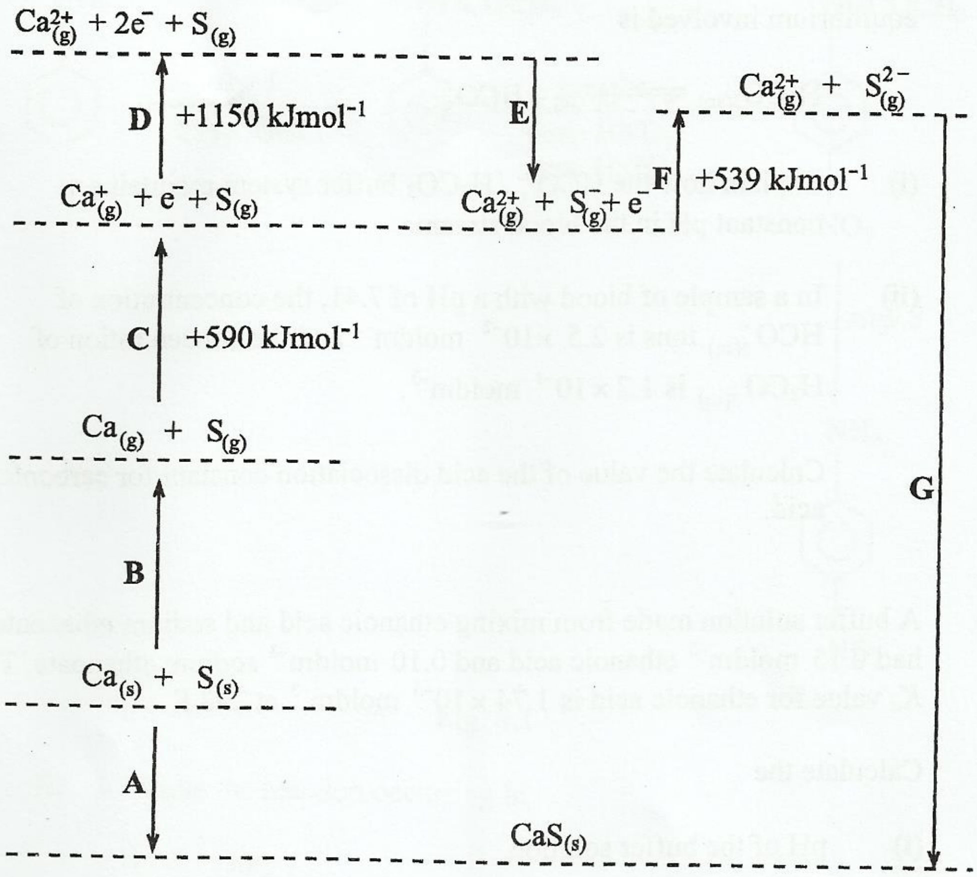


Fig. 2.1

(i) State the electronic configuration of the sulphide ion, S^{2-}.

(ii) Name enthalpy changes **D** and **E**.

(iii) Explain why the enthalpy change for **D** is larger than for **C**.

(iv) Suggest why **F** is an endothermic process.

[6]

(b) When a 0.218 mole sample of hydrogen iodide was heated in a flask of volume, V dm^3, the equilibrium established at 700 K was

$$2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}.$$

The equilibrium mixture was found to contain 0.023 moles of hydrogen.

(i) State and explain the effect of increasing pressure on the equilibrium.

(ii) Calculate the value of K_c at 700 K for the equilibrium.

[6]
[Total: 12]

3 **(a)** In the human body, one important buffer system in the blood involves the hydrogen carbonate ion, HCO_3^-, and carbonic acid, H_2CO_3. The equilibrium involved is

$$H_2CO_{3(aq)} \rightleftharpoons H^+_{(aq)} + HCO^-_{3(aq)}.$$

(i) Explain how the HCO_3^- / H_2CO_3 buffer system maintains a constant pH in the blood stream.

(ii) In a sample of blood with a pH of 7.41, the concentration of $HCO^-_{3(aq)}$ ions is 2.5×10^{-2} $moldm^{-3}$ and the concentration of $H_2CO_{3(aq)}$ is 1.2×10^{-3} $moldm^{-3}$.

Calculate the value of the acid dissociation constant for carbonic acid.

[4]

(b) A buffer solution made from mixing ethanoic acid and sodium ethanoate had 0.15 $moldm^{-3}$ ethanoic acid and 0.10 $moldm^{-3}$ sodium ethanoate. The K_a value for ethanoic acid is 1.74×10^{-5} $moldm^{-3}$ at 298 K.

Calculate the

(i) pH of the buffer solution,

(ii) change in pH on addition of a 10 cm^3 portion of 1.0 $moldm^{-3}$ hydrochloric acid to 1 dm^3 of the buffer solution.

[8]
[Total: 12]

Section B

*Answer at least **one** question from this section.*

4 (a) State and explain the trend in the melting points of Period 2 elements, lithium to neon. [5]

(b) The melting points of sodium oxide and phosphorus (V) oxide are 1 215 °C and 24 °C respectively.

Explain the difference in the melting points of the two oxides. [3]

(c) (i) State the observations made when

1. magnesium is heated in air,

2. a stick of white phosphorus is exposed to air at 40 °C.

(ii) Write the chemical equations for each of the reactions in **(i)**.

[4]
[Total: 12]

5 (a) (i) State the role of chlorine in the treatment of water.

(ii) Give a reason why the amount of chlorine must be carefully monitored in the treatment of water.

(iii) Write an equation for the reaction of chlorine with water.

(iv) State any other **two** uses of chlorine.

[5]

(b) Fumes of hydrogen chloride and hydrogen iodide are separately bubbled through water.

State, with reasons, the halide that gives a solution with a higher pH. [3]

(c) (i) Write a balanced chemical equation for the reaction of chlorine with

1. cold aqueous NaOH,

2. hot aqueous NaOH.

(ii) State the changes in the oxidation states of chlorine for each of the reactions in **(i)**.

[4]
[Total:12]

Section C

Answer at least two questions from this section.

6 **(a)** Organic compounds, **P**, **Q** and **R** are non-cyclic branched hydrocarbons of the formula C_6H_{12}. **P** shows optical isomerism while **Q** and **R** show geometrical isomerism.

Deduce the possible structures of **P**, **Q** and **R**. [3]

(b) **Fig. 6.1** shows the structure of an organic compound **Z**.

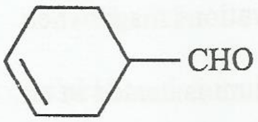


Fig. 6.1

(i) Draw the structures of the organic products obtained when

1. a few drops of Fehlings solution are added to **Z**,

2. **Z** reacts with cold alkaline $KMnO_4$.

(ii) State the observable changes that occur in each of the reactions in (i).

(iii) Describe, with the aid of chemical equations, the changes that occur when hydrogen cyanide is added to **Z** under acidic conditions.

[6]

(c) **Fig. 6.2** shows the structure of a triester.

$$\begin{array}{r}
CH_3-(CH_2)_6-COOCH_2 \\
| \\
CH_3-(CH_2)_6-COOCH \\
| \\
CH_3-(CH_2)_6-COOCH_2
\end{array}$$

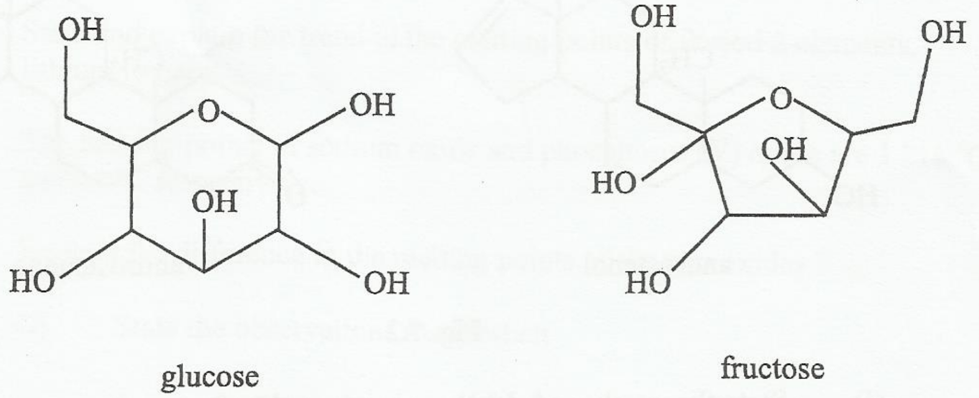
Fig. 6.2

(i) Draw the structures of the organic products formed when the triester is heated under reflux with an excess of aqueous sodium hydroxide.

(ii) Compare the solubilities, in water, of the organic products in (i) and the triester.

[3]
[Total: 12]

7 (a) **Fig. 7.1** shows the structures of glucose, fructose and gluconic acid, the major constituents of honey. Bees make honey by concentrating the sugars and removing excess water in nectar.



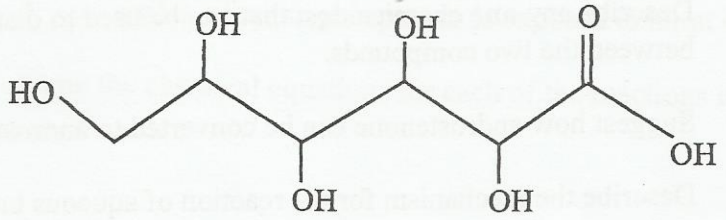


Fig. 7.1

(i) State the molecular formula of

1. glucose,

2. fructose.

(ii) Deduce, from **(i)** the relationship between glucose and fructose.

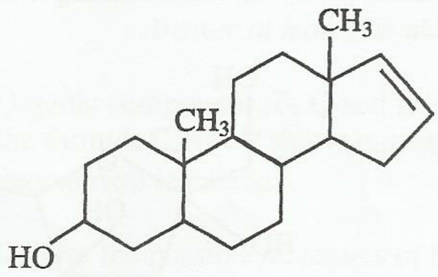
(iii) State, with reasons, the effect of honey on plane polarised light.

(iv) Suggest why honey, when kept at normal temperature, does not get bad over a long period of time.

[6]

[Turn over

(b) **Fig. 7.2** shows the structural formula of androstenol and androstenone.



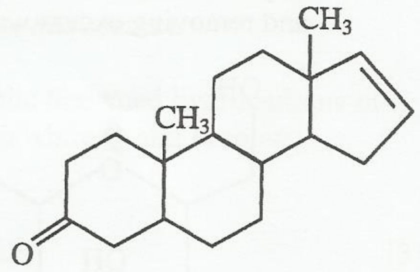


Fig. 7.2

(i) State the number of chiral centres in androstenone.

(ii) Describe any **one** chemical test that can be used to distinguish between the two compounds.

(iii) Suggest how androstenone can be converted to androstenol.

(iv) Describe the mechanism for the reaction of aqueous bromine with androstenone.

[6]
[Total: 12]

8 (a) The reaction scheme, in **Fig. 8.1** shows the synthesis of 4-nitrophenylamine from phenylamine.

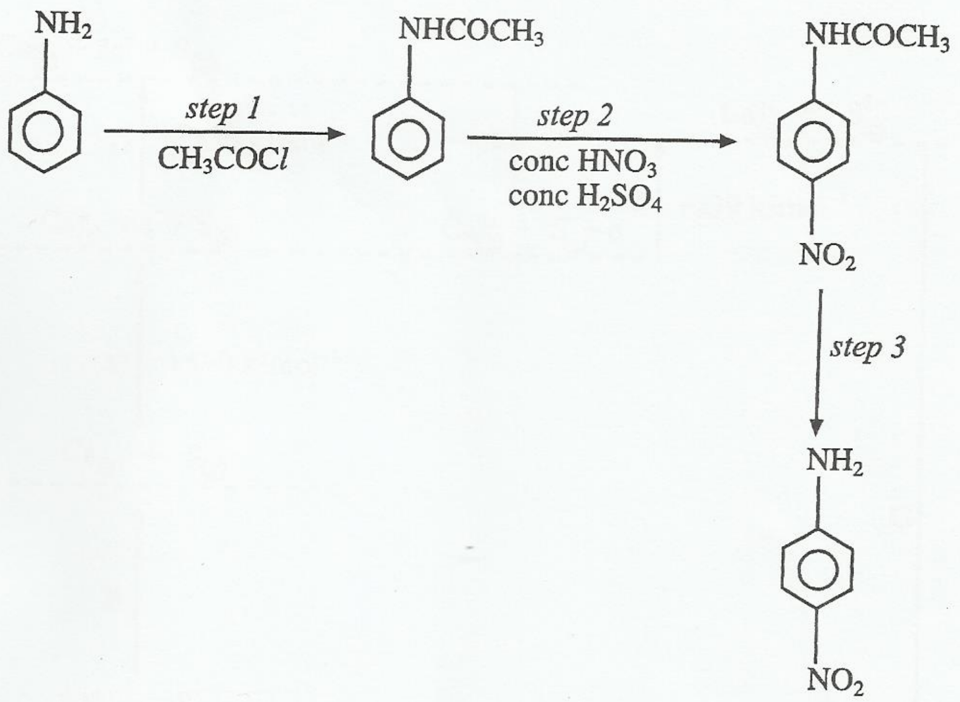


Fig. 8.1

(i) Name the reaction occurring in

1. *step 1*,

2. *step 2*.

(ii) State reagents and conditions for *step 3*.

(iii) Outline the mechanism for the reaction in *step 2*.

[6]

(b) Outline a three step synthesis of the compound, $CH_3CH_2NH_2$ using methane as the starting material.

[5]

(c) Explain why ethylamine is a base. [1]

[Total: 12]